

On one platform, agents collect their own data and train on it, month after month, so the model compounds without human intervention. The system ran for fifteen unattended rounds, and each stage's contribution was measured.

Weed Detection

The corpus exceeds 150,000 labelled images. Two agents operate on it unprompted: one harvests and audits imagery, the other trains a detector small enough for the robots.

Rover

A four-wheel skid-steer the width of a crop row carries a Jetson, a Raspberry Pi and its own battery. Camera, GPS and two IMUs stream live over one API key.

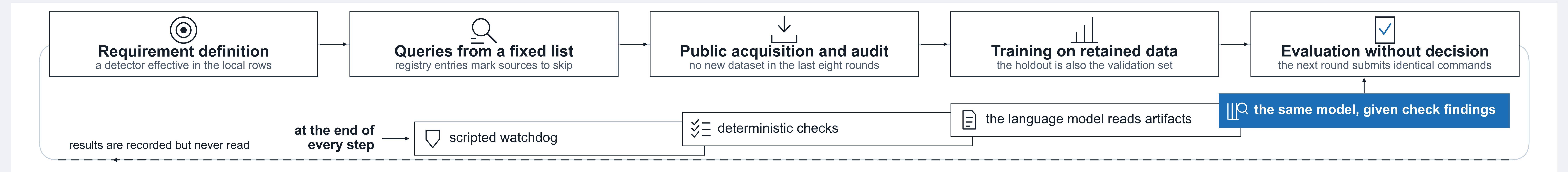
Laser Weed Control Robot

A YOLO detector localises each weed; two galvanometer lasers aim and fire, destroying it in place. No chemical is applied and nothing contacts the adjacent crop.

Humanoid and Quadruped

This project pairs a humanoid with a quadruped to record terrain that wheels do not reach. It adopts the Rover's ingest contract, so every session the robots open is governed.

Figure 1. Every project on the platform is created by the same procedure and registers its data under one ingest contract. The harvesting and training agents have so far run on one of them.



Agent Workflow Diagram

Figure 2. A requirement enters. The harvest issues fixed queries without gap analysis. Audited data train the model; the holdout is also the validation set and selects the reported epoch. Evaluation parses a metric; no decision follows. The reviewer runs after every step; the dashed edge is unimplemented.

ONE PLATFORM SUPPORTS ANY ROBOT OR DATASET

agentAI is one platform for every laboratory robot

A robot, drone or sensor rig is a project with its own keys, members and access-controlled dataset registry. Agents harvest, filter, label and train against that registry. Adding a robot is a configuration change.

Both the Rover and the laser robot transmit data

The Rover is a four-wheel skid-steer streaming camera, GPS and two inertial measurement units (IMUs). The Laser Weed Control Robot carries two galvanometer lasers on independent channels, a downward camera over the work zone and a forward camera.

Each drive becomes a dataset at its start

The platform registers a vehicle session as a governed dataset during the drive. Frames and sensor batches arrive over one API key. The laser robot transmits no position.

Every data type and domain forms its own project

On upload, a project returns an automatic diagnosis of the data before the next step is chosen. Weed detection is one project; the Rover and the laser robot are two others.



The platform is live; the QR code links to it

lab-b660m-c.tailfa6424.ts.net

Authentication requires an institutional Google account.

OPEN-WEIGHT MODELS REVIEW IN TIERS

Grounded recall distinguishes the reviewers

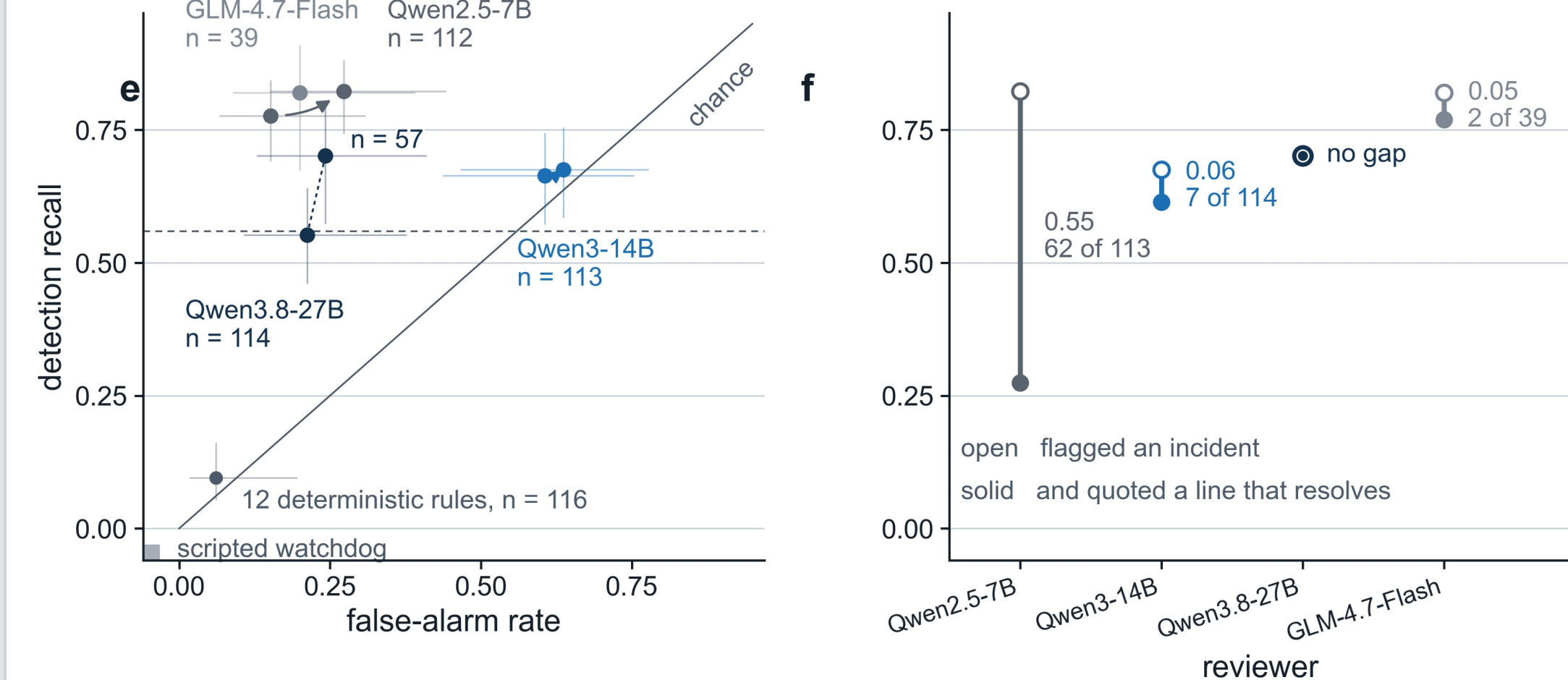


Figure 3. (e) Recall and (f) grounded recall on one frozen corpus of real incidents from this project's record, scored by its own scorer. Each tail is the language model on artifacts with the twelve checks' findings withheld; each head is the same model supplied with them. The dotted line has no arrowhead: its ends were scored on different cases, the 27 B head on 57 incidents and its tail on 114.

Every reviewer is an open-weight model served from a self-hosted cluster. The higher tier adds grounding, not parameter count: supplying the deterministic tier's findings to the model yields +0.149 recall at 27 B and +0.012 at 14 B.

Tiered review is invoked at the end of every step

The tiers are a scripted watchdog, twelve deterministic checks, a language model reading the artifacts with those checks withheld, then the same model supplied with them. On 149 frozen cases the scripted watchdog returned no decision.

AGENTS RAN THE LOOP WITHOUT AN OPERATOR

Compounding was the premise of the unattended loop

One agent collects and one trains, so each round's model should exceed the last. Fifteen rounds ran unattended. The training corpus did not change after round three, yet the ledger and every round report recorded each round as new work.

MORE DATA NO LONGER IMPROVED THE MODEL

More harvested data measurably reduced accuracy

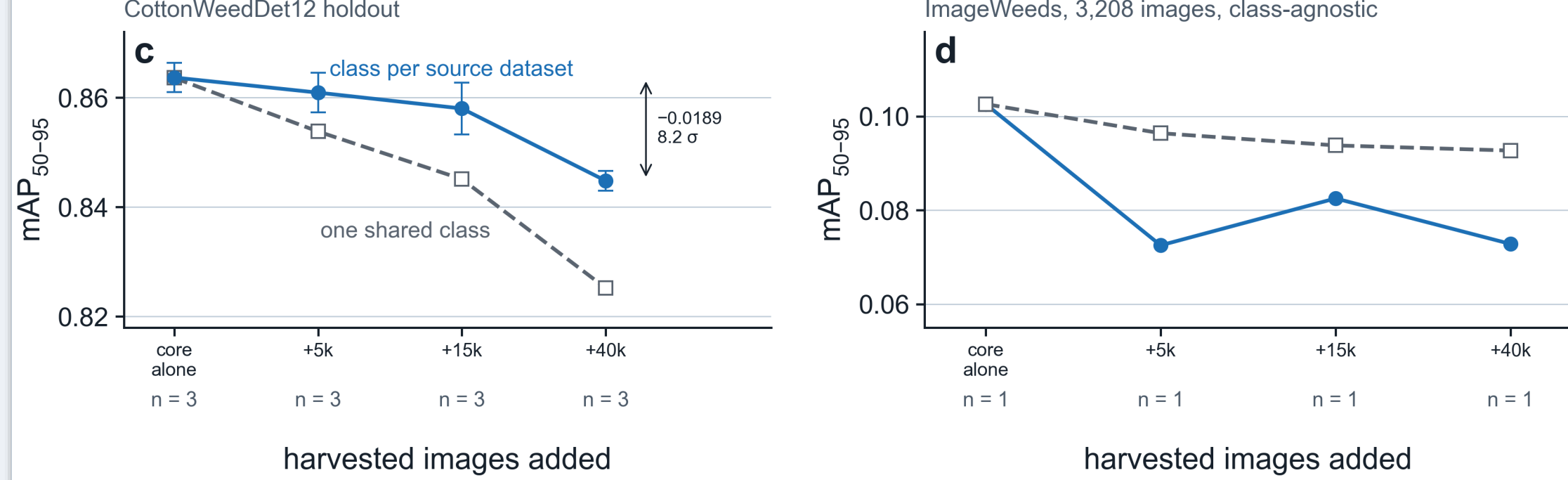


Figure 4. (c) Three seeds at each level, mean and 1 s.d., on the 1,977-image holdout, which is also each run's validation set, so every mark is that run's best epoch. (d) The same eight checkpoints on an unseen weed dataset, one seed per level. Both panels decline.

The target is 0.90 mAP50-95, above the 0.869 published for this class of task. Re-initialising training rather than resuming from the previous round's weights recovers +0.0287, five times the seed spread of this recipe.

MORE DATA NO LONGER IMPROVED THE MODEL (CONT.)

Training configuration was not the limiting factor

Data continued to arrive; the detector did not improve. Augmentation, replaying known species and completing labels each left the same gap. The explanation that remains is the automatic labeller's boxes.

Geometry rejects boxes the language model accepts

The rule discards boxes lying too far from the trusted data. Presented with a candidate box, Gemma 4 rejects none of the false boxes, so the criterion is DINOv2 geometry. Fifty-one datasets were removed; retraining did not restore accuracy.

The unit of curation must match the unit of noise

An automatic-labelling error is a single incorrect box in an otherwise acceptable image, so a dataset-level filter cannot detect it. The per-box tools are implemented but not re-measured. That remains an open question.

A language model's falsifiable hypothesis was false

Harvested boxes take their class from the source website. A language model proposed that as the cause, stated its falsifying condition first, and a human approved. Four runs falsified it. The cost was density: nine boxes per image against under two.